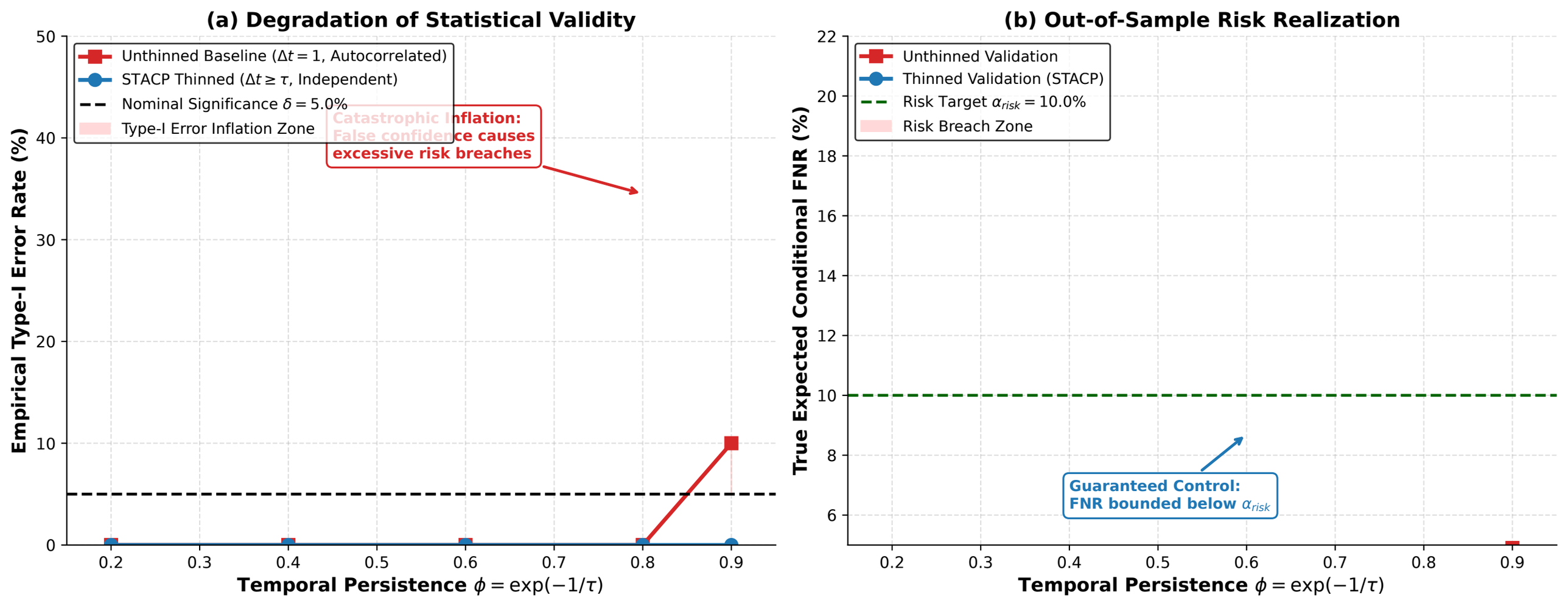
(a) Degradation of Statistical Validity
Empirical Type-I Error Rate (%)
Temporal Persistence $\phi = \exp(-1/\tau)$
Unthinned Baseline ($\Delta t = 1$, Autocorrelated)
STACP Thinned ($\Delta t \geq \tau$, Independent)
Nominal Significance $\delta = 5.0\%$
Type-I Error Inflation Zone
Catastrophic Inflation:
False confidence causes
excessive risk breaches
(b) Out-of-Sample Risk Realization
True Expected Conditional FNR (%)
Temporal Persistence $\phi = \exp(-1/\tau)$
Unthinned Validation
Thinned Validation (STACP)
Risk Target $\alpha_{risk} = 10.0\%$
Risk Breach Zone
Guaranteed Control:
FNR bounded below α_{risk}